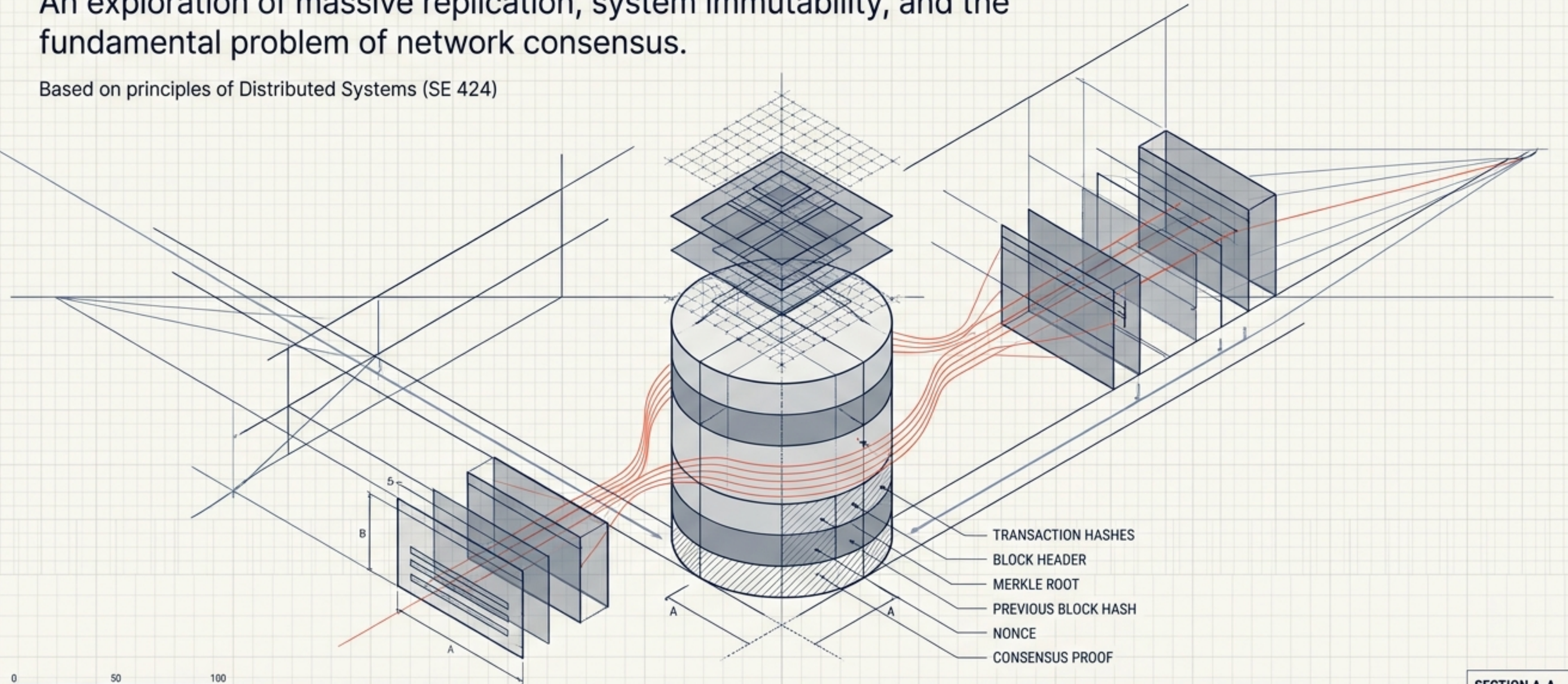


Mapping the Architecture of Distributed Ledgers

An exploration of massive replication, system immutability, and the fundamental problem of network consensus.

Based on principles of Distributed Systems (SE 424)

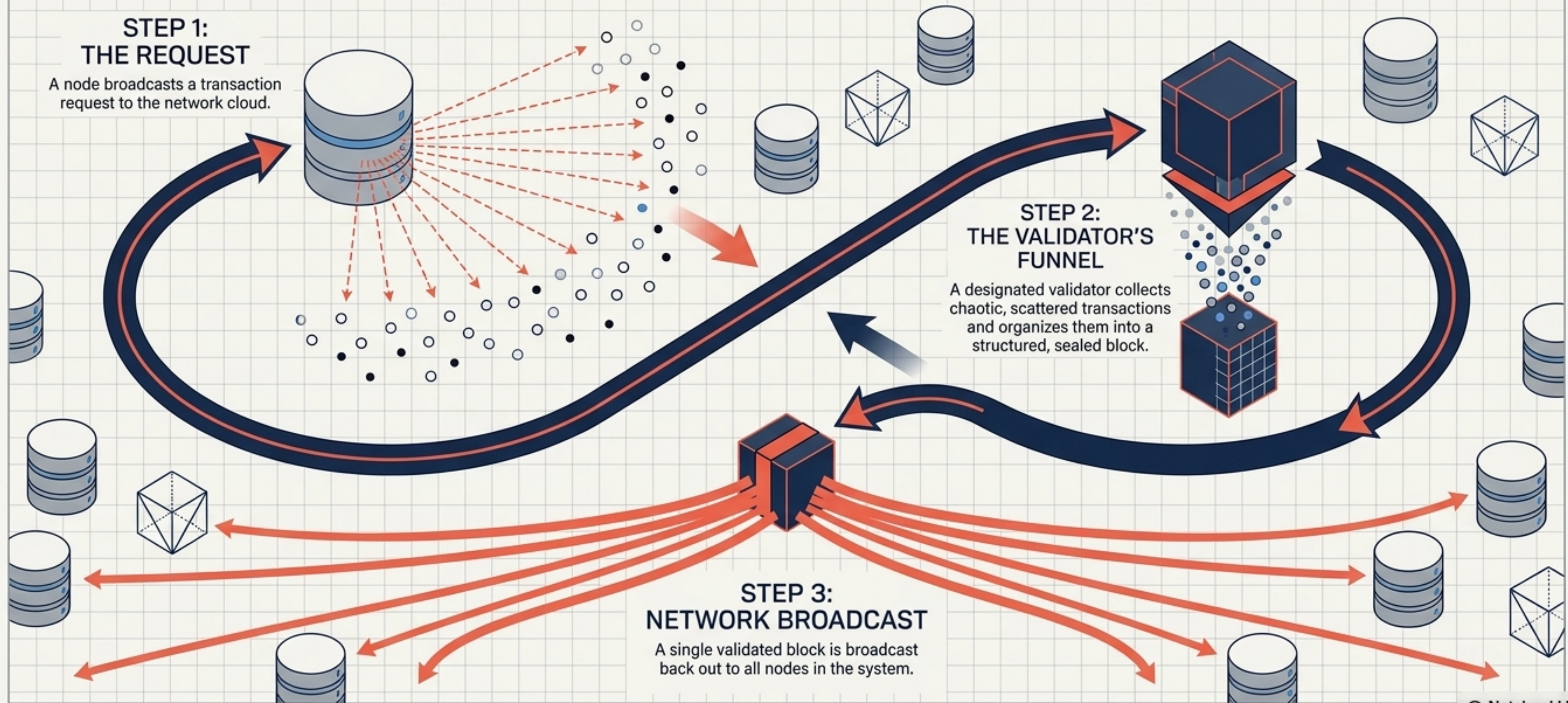


SECTION A-A
SCALE 1:100

APPROVED: ARCHITECTURAL DISTRIBUTED SYSTEMS TEAM

THE INFINITE CONSENSUS LOOP

Blocks are organized into an unforgeable, append-only chain.



THE IMMUTABILITY MANDATE REQUIRES MASSIVE REPLICATION

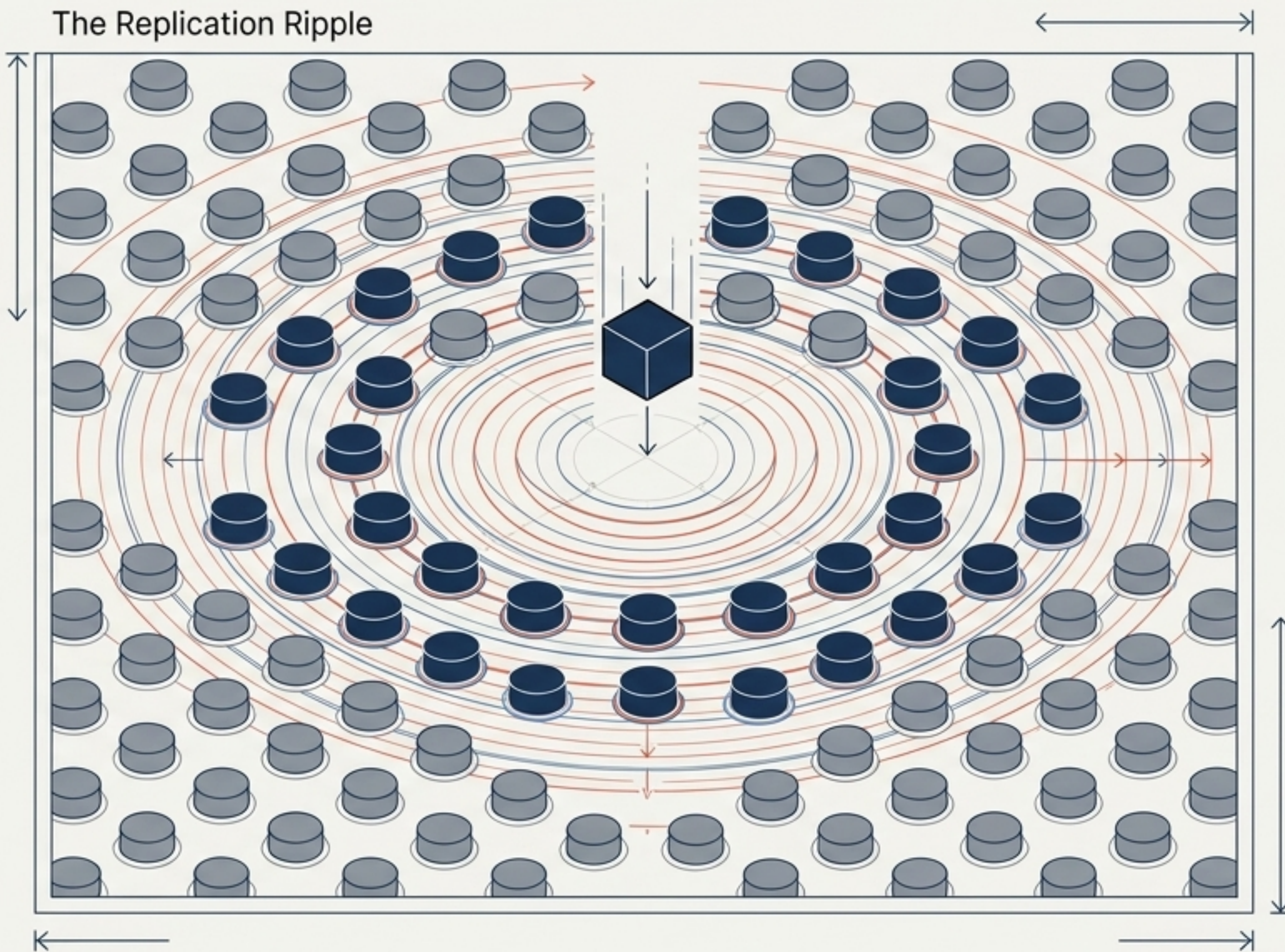
THE RULE: EACH BLOCK IN THE BLOCKCHAIN IS IMMUTABLE.

Once a block is cryptographically sealed and added to the chain, its contents, including all transactions and metadata, cannot be changed, edited, or deleted.

THE CONSEQUENCE: BECAUSE THE LEDGER CANNOT BE ALTERED AFTER THE FACT, THE SYSTEM RELIES ON MASSIVE REPLICATION ACROSS THE ENTIRE NODE ECOSYSTEM TO MAINTAIN TRUTH.

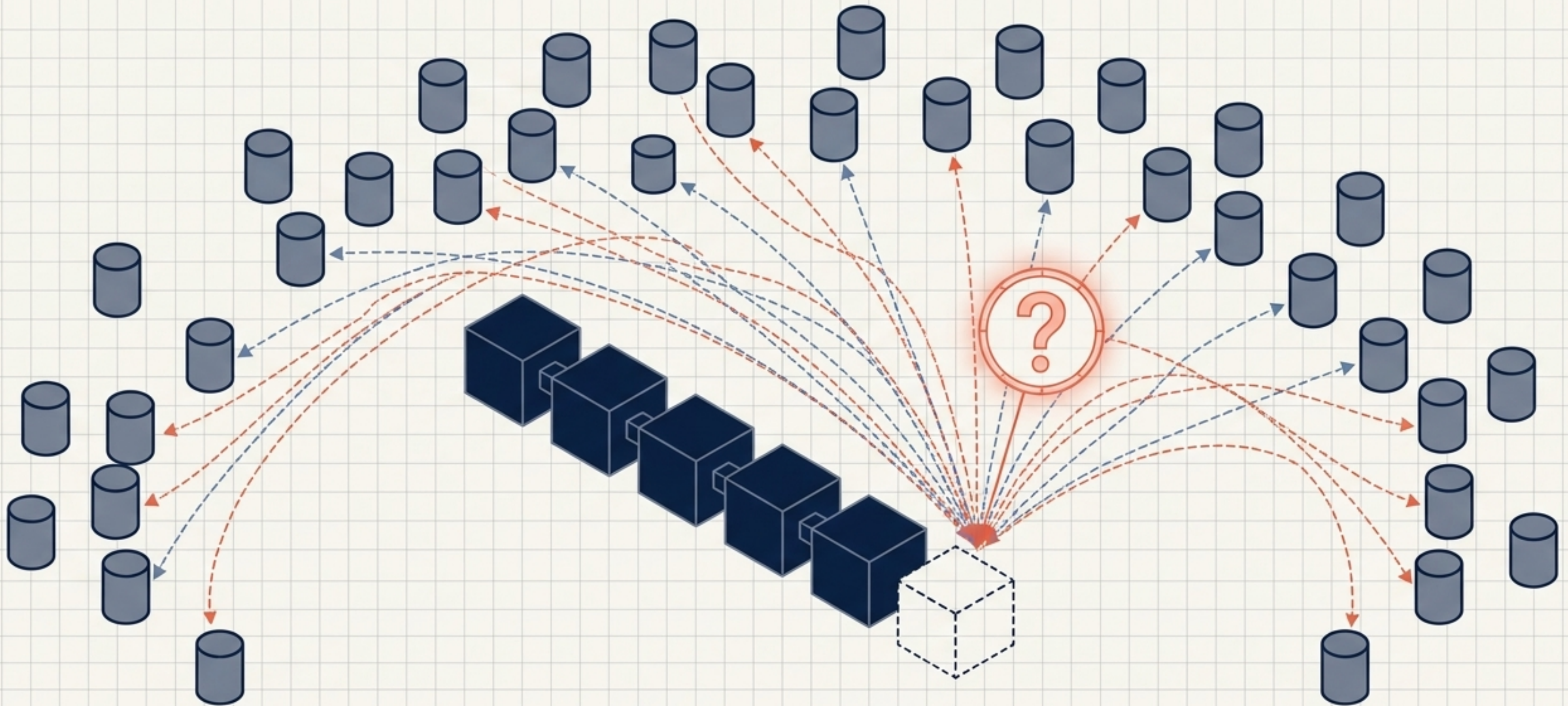
To ensure resilience and prevent data loss, the entire history of the blockchain must be independently stored and continuously validated by every full node in the network, leading to extensive redundancy.

The Replication Ripple

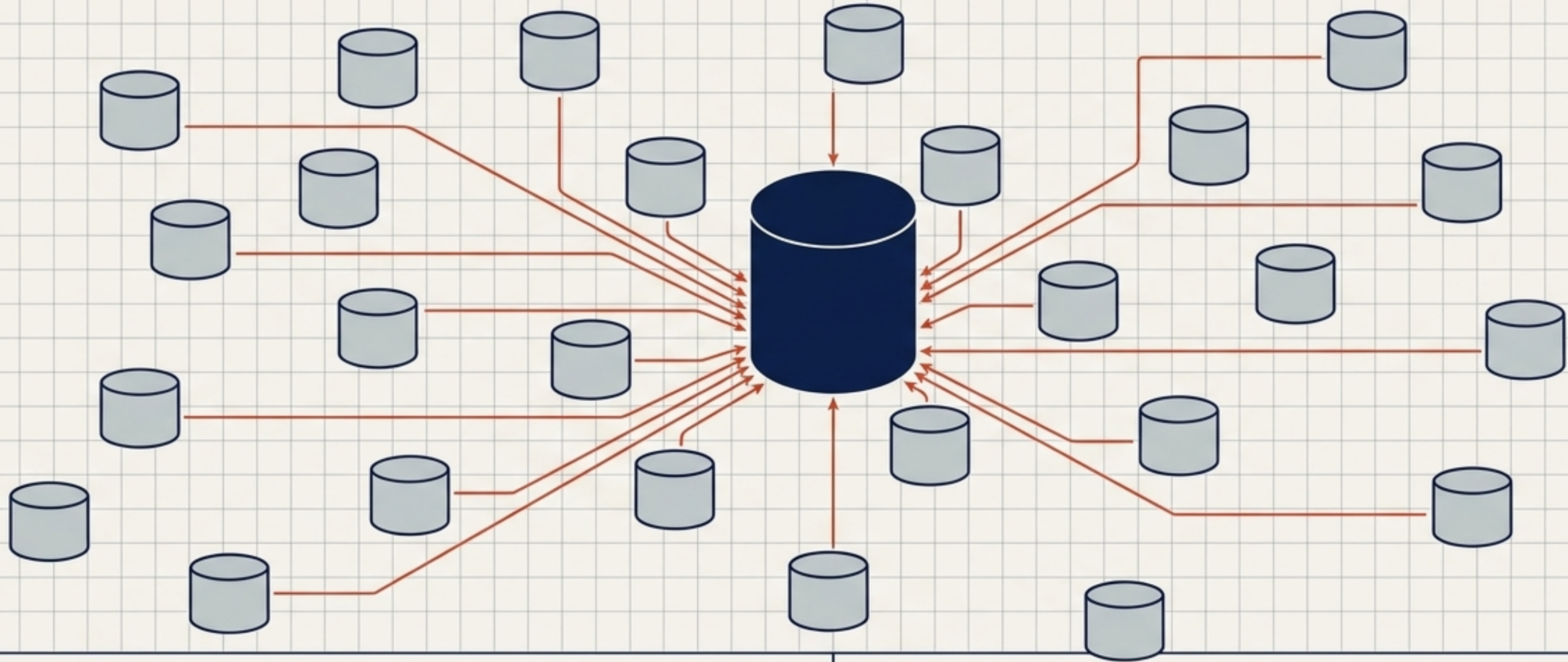


THE FUNDAMENTAL SNAG IN DISTRIBUTED APPEND SYSTEMS

The Core Conflict: The real snag lies in who is allowed to append a block to a chain.



CENTRALIZED AUTHORITY DEFEATS THE SYSTEM'S DESIGN GOALS



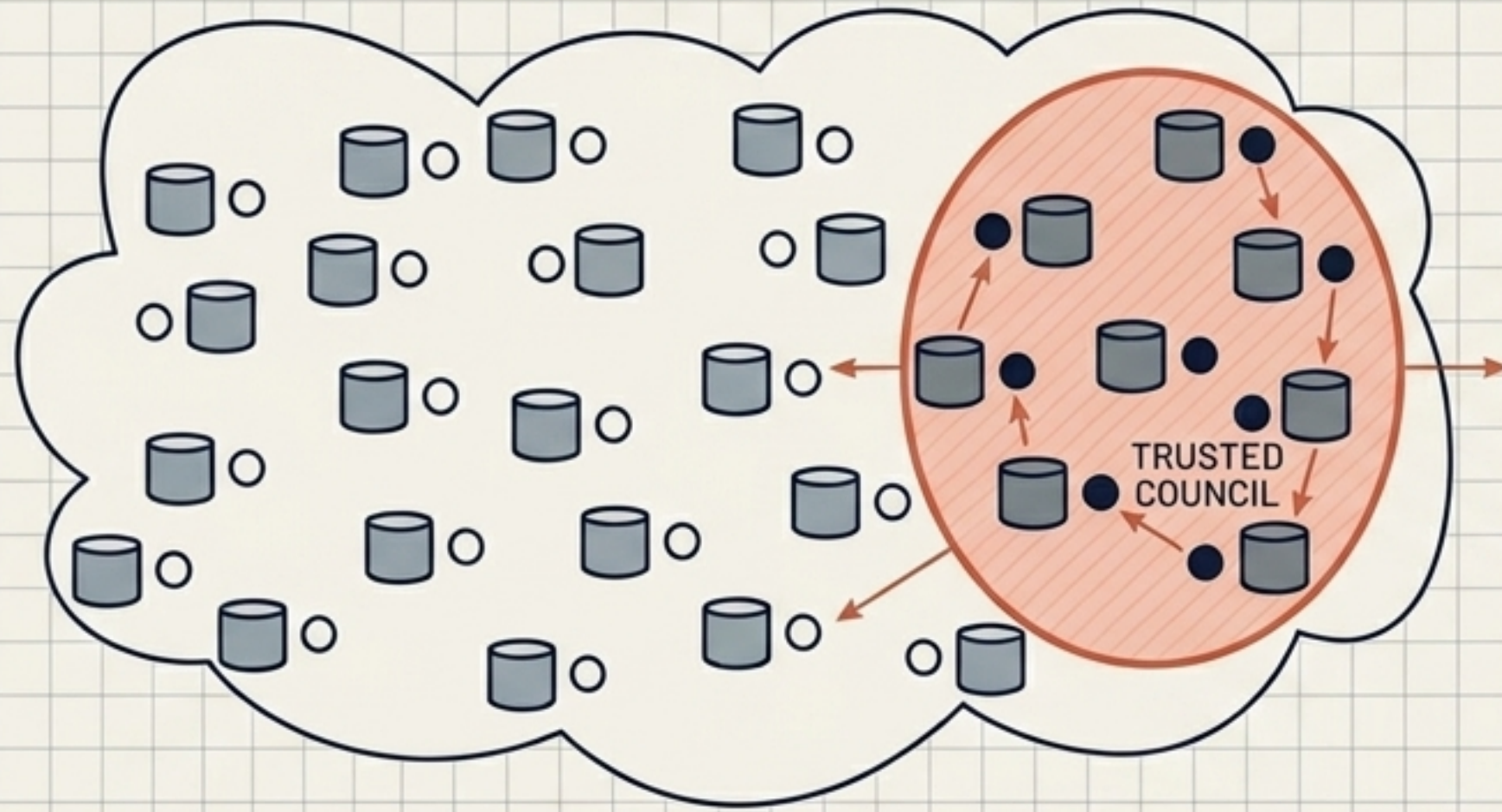
THE MECHANISM

A single entity decides which validator can go ahead and append a block.

THE VERDICT

Does not fit the design goals of blockchains.

DISTRIBUTED PERMISSIONED NETWORKS RELY ON TRUSTED COUNCILS



The Mechanism: A selected, relatively small group of servers jointly reach consensus on which validator can go ahead.

The Scale Limit: In practice, only 10s of servers can be accommodated in this architecture.

THE TRUST MATH: $> 2/3$



None of the individual servers needs to be trusted natively, as long as roughly $2/3$ behave according to system specification.

DISTRIBUTED PERMISSIONLESS NETWORKS REQUIRE COMPLEX LOTTERIES

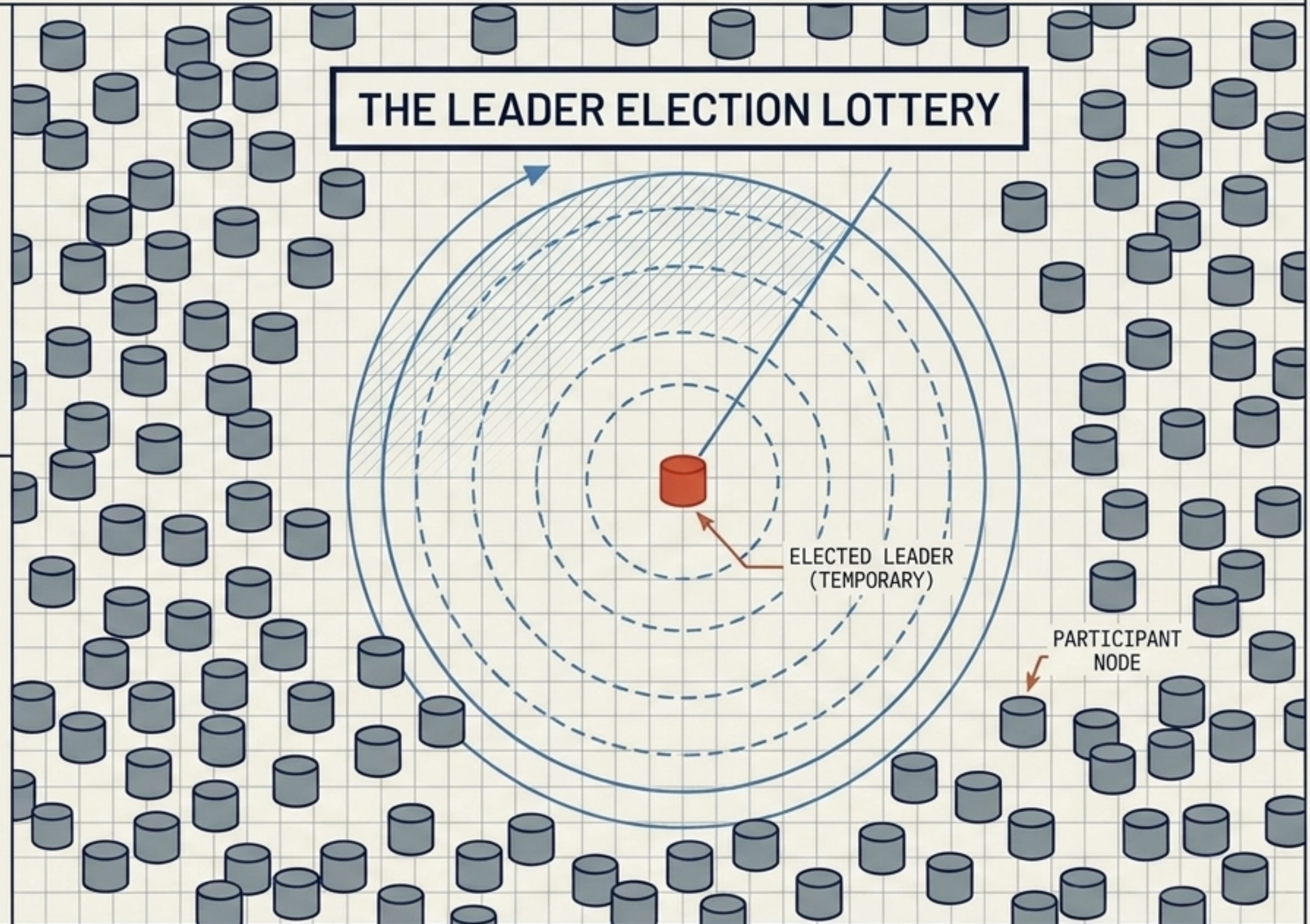
THE MECHANISM

Participants collectively engage in a leader election.

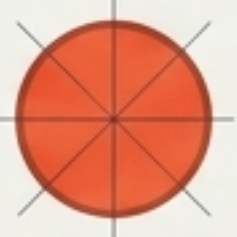
Only the elected leader is allowed to append a block of valid transactions.

THE CHALLENGE

Designing a large-scale, decentralized leader election that remains fair, robust, and secure is highly non-trivial.

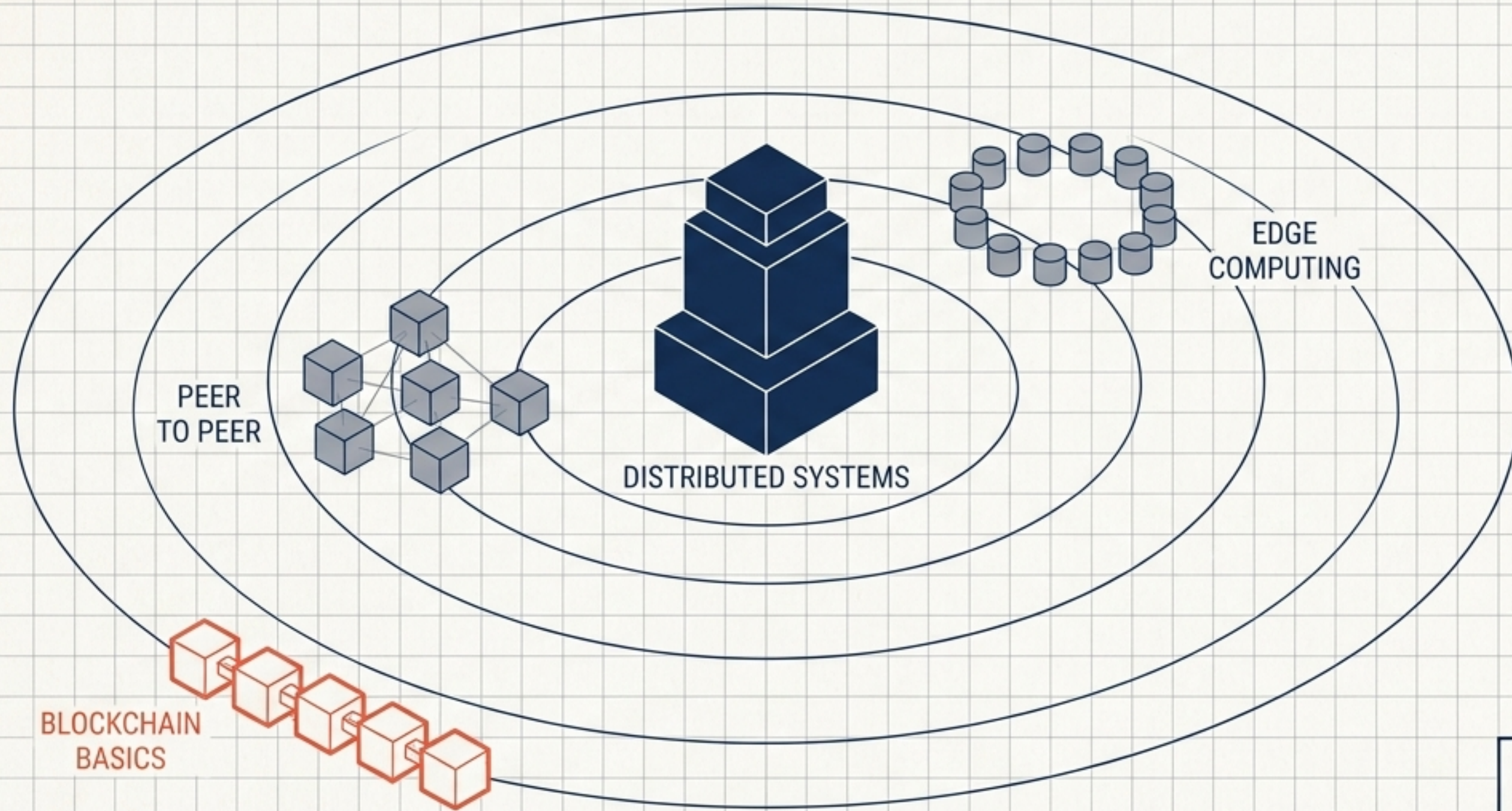


THE CONSENSUS ARCHITECTURE MATRIX

	CENTRALIZED	DISTRIBUTED PERMISSIONED	DISTRIBUTED PERMISSIONLESS
APPENDING AUTHORITY	Single Entity	Small Selected Group	Elected Leader
TRUST ASSUMPTION	Trust the Center	Trust $\approx 2/3$ of Group	Trust the Election Protocol
SCALE LIMITS	Unlimited nodes, 1 decider	10s of Servers	Large-scale, Decentralized
BLOCKCHAIN ETHOS FIT	Fails 	Compromise 	Aligns 

CONTEXTUALIZING BLOCKCHAIN WITHIN DISTRIBUTED ARCHITECTURES

Blockchain consensus is just one paradigm within the broader architectural landscape of modern distributed networks.



System Architectures:
- Peer to Peer
- Edge Computing
- Blockchain Basics